

Entity Attributes

Primary
Key

Customer	CustomerID, FirstName, LastName, Email, PhoneNumber, Address, City, State, ZipCode, RegistrationDate	CustomerID
Product	ProductID, ProductName, Description, Price, StockQuantity, CategoryID, SupplierID	ProductID
Category	CategoryID, CategoryName, CategoryDescription	CategoryID
Supplier	SupplierID, SupplierName, ContactPerson, Email, PhoneNumber, Address	SupplierID
Order	OrderID, CustomerID, OrderDate, OrderStatus, TotalAmount	OrderID
Payment	PaymentID, OrderID, PaymentDate, PaymentMethod, PaymentAmount, PaymentStatus	PaymentID
Shipment	ShipmentID, OrderID, ShipmentDate, DeliveryDate, TrackingNumber, ShipmentStatus	ShipmentID
Review	ReviewID, CustomerID, ProductID, Rating, ReviewComment, ReviewDate	ReviewID

Foreign Keys

- **CategoryID** in Product → Category(CategoryID)
- **SupplierID** in Product → Supplier(SupplierID)
- **CustomerID** in Order → Customer(CustomerID)
- **OrderID** in Payment → Order(OrderID)
- **OrderID** in Shipment → Order(OrderID)
- **CustomerID** in Review → Customer(CustomerID)
- **ProductID** in Review → Product(ProductID)

Entity Relationship Summary

- One Customer can place many Orders.
- One Order can contain many Products.
- One Category can have many Products.
- One Supplier can supply many Products.
- One Order has one Payment and one Shipment.
- One Customer can write many Reviews.
- One Product can receive many Reviews.

Relationship	Cardinality
Customer places Order	1 : Many
Customer owns Cart	1 : 1
Customer writes Review	1 : Many
Category contains Product	1 : Many
Supplier supplies Product	1 : Many
Order contains Order_Item	1 : Many
Product appears in Order_Item	1 : Many
Order has Payment	1 : 1
Order has Shipment	1 : 1
Product receives Review	1 : Many
Cart contains Cart_Item	1 : Many
Product appears in Cart_Item	1 : Many

Customer

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| Places

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Order ----- Payment

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Contains

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Order_Item ----- Product ----- Category

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Review

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Customer

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Supplier

Customer ----- Cart ----- Cart_Item ----- Product

Order ----- Shipment

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Foreign Keys

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- **SupplierID** in Product → Supplier(SupplierID)
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Customer

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| Places

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Order ----- Payment

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Contains

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Order_Item ----- Product ----- Category

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| Review

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| Customer

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Supplier

Customer ----- Cart ----- Cart_Item ----- Product

Order ----- Shipment